$$\frac{\partial u}{\partial x}+\frac{\partial \eta}{\partial y}\frac{\partial v}{\partial \eta}=0$$

$$u\frac{\partial u}{\partial x}+\mathrm{v}\frac{\partial \eta}{\partial y}\frac{\partial u}{\partial \eta}=\frac{1}{ReD}\frac{\partial \eta}{\partial y}\frac{\partial}{\partial \eta}\left(v\frac{\partial u}{\partial y}\right)$$

The equations are finally reconstructed as follows:

$$\frac{\partial u}{\partial x}+\frac{sy_{max}\cosh(s\eta)}{\sinh(s)}\frac{\partial v}{\partial \eta}=0$$

$$u\frac{\partial u}{\partial x}+\left[v\frac{\sinh(s)}{sy_{max}\cosh(s\eta)}\boldsymbol{\delta}+\frac{\nu}{ReD}\frac{\sinh^2(s)\sinh(s\eta)}{sy_{max}^2\cosh^3(s\eta)}\boldsymbol{\delta^2}\right]\frac{\partial u}{\partial \eta}+\left[\frac{-\nu}{ReD}\frac{\sinh^2(s)\sinh(s\eta)}{s^2y_{max}^2\cosh^2(s\eta)}\boldsymbol{\delta^2}\right]\frac{\partial^2 u}{\partial \eta^2}=0$$

Note that the variable $\boldsymbol{\delta}$ and $\boldsymbol{\delta^2}$ should be included to make the dimension of equations to cancel the dimension of $\mathbf{Y_{max}}$ and $\boldsymbol{Y_{max}^2}$!!

Then, we finally have simplified momentum equation.

Momentum equation to solve U-velocity is discretized as follows: Blue and Red indicate known and unknown value, respectively.

$$u_j^i\left(\frac{u_j^{i+1}-u_j^i}{\Delta x}\right)+\frac{AA}{2}\left[\left(\frac{\partial u}{\partial \eta}\right)_{i,j}+\left(\frac{\partial u}{\partial \eta}\right)_{i+1,j}\right]+\frac{B}{2}\left[\left(\frac{\partial^2 u}{\partial \eta^2}\right)_{i,j}+\left(\frac{\partial^2 u}{\partial \eta^2}\right)_{i+1,j}\right]=0$$

$$\left(\frac{\partial u}{\partial \eta}\right)_{i,j}=\frac{u_{j-2}^i-8u_{j-1}^i+8u_{j+1}^i-u_{j+2}^i}{12\Delta\eta} \qquad \left(\frac{\partial^2 u}{\partial \eta^2}\right)_{i,j}=\frac{-u_{j-2}^i+16u_{j-1}^i-30u_j^i+16u_{j+1}^i-u_{j+2}^i}{12\Delta\eta^2}$$

Final form:

$$(\phi_1-\phi_2)u_{j-2}^{i+1}+(-8\phi_1+16\phi_2)u_{j-1}^{i+1}\left(\frac{u_{i,j}}{\Delta x}-30\phi_2\right)u_j^{i+1}+(8\phi_1+16\phi_2)u_{j\ 1}^{i+1}+(-\phi_1-\phi_2)u_{j+2}^{i+1}$$
$$=(-\phi_1+\phi_2)u_{j-2}^i+(8\phi_1-16\phi_2)u_{j-1}^i+\left(\frac{u_{i,j}}{\Delta x}+30\phi_2\right)u_j^i+(-8\phi_1-16\phi_2)u_{j+1}^i+(\phi_1+\phi_2)u_{j+2}^i$$

$$where\ \phi_1=\frac{AA}{24\Delta\eta} \qquad \phi_2=\frac{B}{24\Delta\eta^2}$$

Near the wall, we need to 2nd order scheme as follows: (j = 2 or JJ-1)

$$u_j^i\left(\frac{u_j^{i+1}-u_j^i}{\Delta x}\right)+\frac{AA}{2}\left[\left(\frac{\partial u}{\partial \eta}\right)_{i,j}+\left(\frac{\partial u}{\partial \eta}\right)_{i+1,j}\right]+\frac{B}{2}\left[\left(\frac{\partial^2 u}{\partial \eta^2}\right)_{i,j}+\left(\frac{\partial^2 u}{\partial \eta^2}\right)_{i+1,j}\right]=0$$

$$\left(\frac{\partial u}{\partial \eta}\right)_{i,j}=\frac{u_{j+1}^i-u_{j-1}^i}{2\Delta\eta} \qquad \left(\frac{\partial^2 u}{\partial \eta^2}\right)_{i,j}=\frac{u_{j+1}^i-2u_j^i+u_{j-1}^i}{\Delta\eta^2}$$

Finally, we have

$$\left(-\frac{AA}{4\Delta\eta}+\frac{B}{2\Delta\eta^2}\right)u_{j-1}^{i+1}+\left(\frac{u_j^i}{\Delta x}-\frac{B}{\Delta\eta^2}\right)u_j^{i+1}+\left(\frac{AA}{4\Delta\eta}+\frac{B}{2\Delta\eta^2}\right)u_{j\ 1}^{i+1}=(\phi_1-\phi_2)u_{j-1}^i+\left(\frac{u_j^i}{\Delta x}+\frac{B}{\Delta\eta^2}\right)u_j^i+\left(-\frac{AA}{4\Delta\eta}-\frac{B}{2\Delta\eta^2}\right)u_{j+1}^i$$

Once we obtain U-velocity of next step, we can obtain V-velocity by solving continuity equation.

4th order derivative for ∂v/ ∂η & 2nd order for ∂u/ ∂x interior domain **(J=3 ~NY)** are as follows:

$$\left(\frac{\partial u}{\partial x}\right)_{j-\frac{1}{2}}^{i+1} = 0.5\left(\frac{3u_j^{i+1} - 4u_j^i + u_j^{i-1}}{2\Delta x} + \frac{3u_{j-1}^{i+1} - 4u_{j-1}^i + u_{j-1}^{i-1}}{2\Delta x}\right)$$

$$\left(\frac{\partial v}{\partial \eta}\right)_{j-\frac{1}{2}}^{i+1} = \frac{v_{j-2}^{i+1} - 27v_{j-1}^{i+1} + 27v_j^{i+1} - v_{j+1}^{i+1}}{24\Delta\eta}$$

$$\left(\frac{\partial v}{\partial \eta}\right)_{j-\frac{1}{2}}^{i+1} = \frac{v_{j-2}^{i+1} - 27v_{j-1}^{i+1} + 27v_j^{i+1} - v_{j+1}^{i+1}}{24\Delta\eta} = -\left(\frac{1}{A_1}\right)_{j-\frac{1}{2}}^{i+1}\frac{0.5}{2\Delta x}\left[3u_j^{i+1} - 4u_j^i + u_j^{i-1} + 3u_{j-1}^{i+1} - 4u_{j-1}^i + u_{j-1}^{i-1}\right]$$

$$\frac{1}{24\Delta\eta}v_{j-2}^{i+1} - \frac{27}{24\Delta\eta}v_{j-1}^{i+1} + \frac{27}{24\Delta\eta}v_j^{i+1} - \frac{1}{24\Delta\eta}v_{j\ 1}^{i+1} = -\left(\frac{1}{A_1}\right)_{j-\frac{1}{2}}^{i+1}\frac{0.5}{2\Delta x}\left[3u_j^{i+1} - 4u_j^i + u_j^{i-1} + 3u_{j-1}^{i+1} - 4u_{j-1}^i + u_{j-1}^{i-1}\right]$$

4th order derivative for ∂v/ ∂η & 1st order for ∂u/ ∂x near the boundary **(i=1 & j=2 and j=NY+1)**

$$\frac{1}{24\Delta\eta}v_{j-2}^{i+1} - \frac{27}{24\Delta\eta}v_{j-1}^{i+1} + \frac{27}{24\Delta\eta}v_j^{i+1} - \frac{1}{24\Delta\eta}v_{j\ 1}^{i+1} = -\left(\frac{1}{A_1}\right)_{j-\frac{1}{2}}^{i+1}\frac{0.5}{\Delta x}\left[u_j^{i+1} - u_j^i + u_{j-1}^{i+1} - u_{j-1}^i\right]$$

- **Calculation of initial value of x**

  To obtain the initial value of X where our initial velocity is given, use the following relationship, $\delta = c\sqrt{(x + x_0)}$, $\delta^2 = c^2(x_0) + c^2(x)$, where δ is boundary layer thickness, and the target is to find x0 **<u>that gives you the correct profile.</u>**